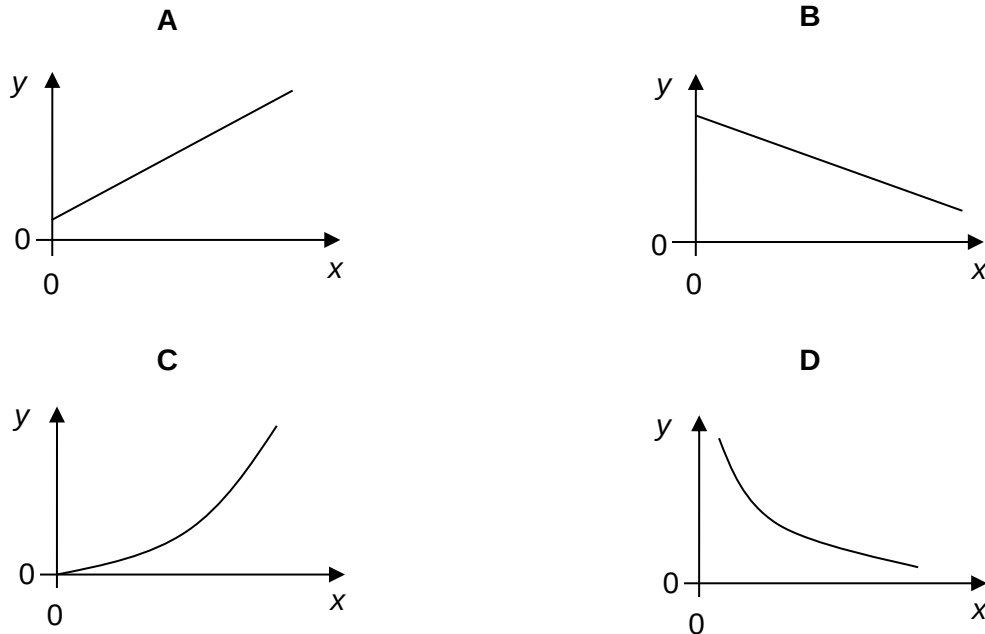


- 1 In an experiment, the perpendicular distance of a point from a long straight conductor carrying a constant current is measured and the perpendicular distance is used to calculate the magnetic flux density due to the long straight current-carrying conductor at that point. The experiment is repeated for a few points.

Which graph shows how the percentage uncertainty in the magnetic flux density of the long straight current-carrying conductor, y , varies with the percentage uncertainty in the perpendicular distance from the conductor, x ?



Answer: A

The magnetic flux density due to a long straight current-carrying conductor is

$$B = \frac{\mu_0 I}{2\pi r}$$

Therefore, the percentage uncertainty of the calculated magnetic flux density is

$$\frac{\Delta B}{B} = \frac{\Delta r}{r} + \frac{\Delta I}{I}$$

Therefore, the graph of $\frac{\Delta B}{B}$ against $\frac{\Delta r}{r}$ is a straight line graph with positive gradient and y -intercept

$$\frac{\Delta I}{I}$$